Vector Integration: Stokes' Theorem on a Quadrant Region

1. Stokes' Theorem Over a Flat 2D Planar Region

Name of Concept

Stokes' Theorem over a Planar Surface bounded by coordinate axes and a circular arc.

Core Mathematical Formulas

Stokes' Theorem general statement:

$$\oint_C \bar{F} \cdot d\bar{r} = \iint_S (\nabla \times \bar{F}) \cdot \hat{n} \, dS$$

For a region lying flat inside the xy-plane ($z = 0$), the standard unit outward normal vector points vertically up:

$$\hat{n} = \bar{k}$$

The Cartesian double-integral area transformation element simplifies to:

$$dS = dx \, dy$$

Beginner-Friendly Explanation of Operations

1. **Defining the Boundaries:** The problem specifies that the boundary curve C lies completely flat on the floor plane ($z = 0$). It forms a wedge or quadrant shape bounded by the y-axis ($x = 0$), the x-axis ($y = 0$), and a circular track of radius 1 ($x^2 + y^2 = 1$). This means our region S is exactly one-quarter of a circle sitting in the first quadrant.
2. **Locking z to the Surface:** Because the entire surface rests flat on the $z = 0$ floor plane, every point across this surface has a vertical coordinate of exactly zero. We can substitute $z = 0$ directly into our computed curl function to simplify it before starting the integration.
3. **The Normal Alignment ($\cdot \, \bar{k}$):** Since the surface is flat on the floor, its directional orientation points straight up along the $\bar{k}$ axis. When we take the dot product $(\nabla \times \bar{F}) \cdot \bar{k}$, it extracts *only* the third component (the $\bar{k}$ part) of the curl vector. Any swirling forces happening sideways along the $\bar{i}$ or $\bar{j}$ directions are ignored because they do not pierce through the floor.

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2. Step-by-Step Problem Solution

Question

Use Stokes' Theorem to evaluate $\oint_C \bar{F} \cdot d\bar{r}$ where $\bar{F} = 4xz\bar{i} - y^2\bar{j} + yz\bar{k}$ and C is the boundary of the area in the plane $z = 0$ bounded by $x = 0, y = 0$, and $x^2 + y^2 = 1$.

Step 1: Identify vector field components

Extract the three individual scalar function parts from the field equation:

$$f_1 = 4xz, \quad f_2 = -y^2, \quad f_3 = yz$$

Step 2: Set up and expand the Curl determinant matrix

$$\nabla \times \bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4xz & -y^2 & yz \end{vmatrix}$$

Evaluate each cross-derivative component carefully:

$$\text{Component } \bar{i} = \left[\frac{\partial}{\partial y}(yz) - \frac{\partial}{\partial z}(-y^2)\right] = z - 0 = z$$

$$\text{Component } \bar{j} = -\left[\frac{\partial}{\partial x}(yz) - \frac{\partial}{\partial z}(4xz)\right] = -(0 - 4x) = 4x$$

$$\text{Component } \bar{k} = \left[\frac{\partial}{\partial x}(-y^2) - \frac{\partial}{\partial y}(4xz)\right] = 0 - 0 = 0$$

Assemble the full rotational vector expression:

$$\nabla \times \bar{F} = z\bar{i} + 4x\bar{j} + 0\bar{k}$$

Step 3: Compute the scalar dot product $(\nabla \times \bar{F}) \cdot \hat{n}$

Substitute the plane orientation unit normal vector $\hat{n} = \bar{k}$:

$$(\nabla \times \bar{F}) \cdot \hat{n} = (z\bar{i} + 4x\bar{j} + 0\bar{k}) \cdot \bar{k}$$

$$(\nabla \times \bar{F}) \cdot \hat{n} = 0$$

Step 4: Apply the plane constraint ($z = 0$)

The dot product calculation results in a value of 0 because the $\bar{k}$ component of our specific curl happens to be zero.

(Note: Even if the dot product had retained a variable function containing z, substituting the flat floor constraint $z = 0$ would collapse any remaining z terms to 0 as well).

Step 5: Substitute the final value into Stokes' Theorem template

Assemble the surface integral using our calculated value of zero:

$$\oint_C \bar{F} \cdot d\bar{r} = \iint_S 0 \, dS = 0$$

Because the field has no rotational twist components pushing perpendicularly through the flat surface area, the accumulated work executed around the perimeter path evaluates to zero.

Final Answer

$$\oint_C \bar{F} \cdot d\bar{r} = 0$$